THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Power Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 3^5	Add the exponents on top, then subtract the bottom exponent
2	**(d)** $\frac{9}{4}$	$(a/b)^{-n} = (b/a)^n$
3	**(c)** 7	$x + 4 = 11$
4	**(a)** 4	$3x = 12$
5	**(c)** 1000	$a^m \times b^m = (ab)^m = (10)^3$
6	**(d)** 6×10^7	Multiply the numbers, add the exponents, then adjust
7	**(c)** 431	$512 - 81 = 431$
8	**(a)** $\frac{5}{6}$	$\frac{1}{2} + \frac{1}{3}$
9	**(b)** 5^4	Add the exponents on top, then subtract the bottom exponent
10	**(d)** $\frac{25}{9}$	$(a/b)^{-n} = (b/a)^n$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$(2^3)^3 = 2^9$, so $2^9 \div 2^5 = 2^4$ (1) $= 16$ (1).	1+1
12	$\frac{1}{32} = 2^{-5}$ (1); so $x = -5$ (1).	1+1
13	$3 \times 10^{8-3}$ (1) = 3×10^5 (1).	1+1
14	$(3^2)^4 = 3^8$, so $3^8 \div 3^5 = 3^3$ (1) $= 27$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$2^2 \times 3^2$ (1+1); value $4 \times 9 = 36$ (1).	1+1+1
16	$\frac{1}{8} + \frac{1}{9}$ (1); common denominator (1); sum $\frac{17}{72}$ (1).	1+1+1
17	Total $= 2.5 \times 10^4 \times 12$ (1) $= 30 \times 10^4$ (1); in scientific notation 3×10^5 (1).	1+1+1
	OR: $3^2 \times 2^3$ (1+1); value $9 \times 8 = 72$ (1).	1+1+1
18	$\frac{1}{16} + \frac{1}{25}$ (1); common denominator (1); sum $\frac{41}{400}$ (1).	1+1+1
	OR: Total $= 1.5 \times 10^3 \times 40$ (1) $= 60 \times 10^3$ (1); in scientific notation 6×10^4 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Numerator power of 2: $2^3 \div 2^{-2} = 2^5$ (2); for 3: $3^{-1} \div 3^1 = 3^{-2} = \frac{1}{3^2}$ (1); result $\frac{2^5}{3^2}$ (1); value $\frac{32}{9}$ (1).	2+1+1+1
	OR: Number of cells $= \frac{8 \times 10^1}{2 \times 10^{-5}}$ (2); $= \frac{8}{2} \times 10^{1+5}$ (2); 4×10^6 (1).	2+2+1
20	Numerator power of 3: $3^2 \div 3^{-1} = 3^3$ (2); for 2: $2^{-1} \div 2^2 = 2^{-3} = \frac{1}{2^3}$ (1); result $\frac{3^3}{2^3}$ (1); value $\frac{27}{8}$ (1).	2+1+1+1
	OR: Number of cells $= \frac{12 \times 10^2}{4 \times 10^{-6}}$ (2); $= \frac{12}{4} \times 10^{2+6}$ (2); 3×10^8 (1).	2+2+1

Mistakes to avoid:

- Writing $2^{-3} = -8$ or $-\frac{1}{8}$; a negative exponent means the **reciprocal**, so $2^{-3} = \frac{1}{8}$.
- Adding the bases: $2^3 \times 2^4 = 4^7$ instead of $2^{3+4} = 2^7$.
- Saying $a^0 = 0$; for $a \neq 0$ it is 1.
- Multiplying exponents when multiplying powers of the same base ($a^m \times a^n = a^{mn}$) or adding them in $(a^m)^n$.
- Leaving the answer as 25×10^4 or 0.25×10^6; in standard form the number part must satisfy $1 \leq m < 10$.
- Getting the sign of the exponent wrong for small numbers: $0.0003 = 3 \times 10^{-4}$, not 3×10^4.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Power Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 7	$x + 2 = 9$
2	**(b)** 5	$4x = 20$
3	**(a)** 10000	$a^m \times b^m = (ab)^m = (100)^2$
4	**(a)** 6×10^7	Multiply the numbers, add the exponents, then adjust
5	**(c)** 17	$81 - 64 = 17$
6	**(c)** $\frac{7}{12}$	$\frac{1}{3} + \frac{1}{4}$
7	**(a)** 2^7	Add the exponents on top, then subtract the bottom exponent
8	**(a)** $\frac{27}{64}$	$(a/b)^{-n} = (b/a)^n$
9	**(c)** 9	$x + 6 = 15$
10	**(b)** 7	$5x = 35$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{1}{27} = 3^{-3}$ (1); so $x = -3$ (1).	1+1
12	$3 \times 10^{-2-2}$ (1) = 3×10^{-4} (1).	1+1
13	$(5^2)^3 = 5^6$, so $5^6 \div 5^4 = 5^2$ (1) $= 25$ (1).	1+1
14	$\frac{1}{25} = 5^{-2}$ (1); so $x = -2$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$5^2 \times 3^2$ (1+1); value $25 \times 9 = 225$ (1).	1+1+1
16	$\frac{1}{9} + \frac{1}{32}$ (1); common denominator (1); sum $\frac{41}{288}$ (1).	1+1+1
17	Total $= 4 \times 10^5 \times 30$ (1) $= 120 \times 10^5$ (1); in scientific notation 1.2×10^7 (1).	1+1+1
	OR: $2^3 \times 7^2$ (1+1); value $8 \times 49 = 392$ (1).	1+1+1
18	$\frac{1}{16} + \frac{1}{25}$ (1); common denominator (1); sum $\frac{41}{400}$ (1).	1+1+1
	OR: Total $= 6.4 \times 10^2 \times 25$ (1) $= 160 \times 10^2$ (1); in scientific notation 1.6×10^4 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Numerator power of 5: $5^1 \div 5^{-2} = 5^3$ (2); for 2: $2^{-2} \div 2^1 = 2^{-3} = \frac{1}{2^3}$ (1); result $\frac{5^3}{2^3}$ (1); value $\frac{125}{8}$ (1).	2+1+1+1
	OR: Number of cells $= \dfrac{15 \times 10^1}{5 \times 10^{-4}}$ (2); $= \frac{15}{5} \times 10^{1+4}$ (2); 3×10^5 (1).	2+2+1
20	Numerator power of 2: $2^4 \div 2^{-1} = 2^5$ (2); for 5: $5^{-1} \div 5^1 = 5^{-2} = \frac{1}{5^2}$ (1); result $\frac{2^5}{5^2}$ (1); value $\frac{32}{25}$ (1).	2+1+1+1
	OR: Number of cells $= \dfrac{7.5 \times 10^2}{2.5 \times 10^{-3}}$ (2); $= \frac{7.5}{2.5} \times 10^{2+3}$ (2); 3×10^5 (1).	2+2+1

Mistakes to avoid:

- Writing $2^{-3} = -8$ or $-\frac{1}{8}$; a negative exponent means the **reciprocal**, so $2^{-3} = \frac{1}{8}$.
- Adding the bases: $2^3 \times 2^4 = 4^7$ instead of $2^{3+4} = 2^7$.
- Saying $a^0 = 0$; for $a \neq 0$ it is 1.
- Multiplying exponents when multiplying powers of the same base ($a^m \times a^n = a^{mn}$) or adding them in $(a^m)^n$.
- Leaving the answer as 25×10^4 or 0.25×10^6; in standard form the number part must satisfy $1 \leq m < 10$.
- Getting the sign of the exponent wrong for small numbers: $0.0003 = 3 \times 10^{-4}$, not 3×10^4.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Power Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 900	$a^m \times b^m = (ab)^m = (30)^2$
2	**(d)** 5×10^7	Multiply the numbers, add the exponents, then adjust
3	**(b)** -118	$125 - 243 = -118$
4	**(d)** $\frac{7}{10}$	$\frac{1}{2} + \frac{1}{5}$
5	**(a)** 7^9	Add the exponents on top, then subtract the bottom exponent
6	**(d)** $\frac{8}{125}$	$(a/b)^{-n} = (b/a)^n$
7	**(d)** 5	$x + 3 = 8$
8	**(a)** 9	$2x = 18$
9	**(b)** 900	$a^m \times b^m = (ab)^m = (30)^2$
10	**(c)** 9×10^3	Multiply the numbers, add the exponents, then adjust

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$2 \times 10^{6-(-2)}$ (1) = 2×10^8 (1).	1+1
12	$(4^3)^2 = 4^6$, so $4^6 \div 4^4 = 4^2$ (1) $= 16$ (1).	1+1
13	$\frac{1}{10000} = 10^{-4}$ (1); so $x = -4$ (1).	1+1
14	$3 \times 10^{5-1}$ (1) = 3×10^4 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$3^3 \times 4^2$ (1+1); value $27 \times 16 = 432$ (1).	1+1+1
16	$\frac{1}{36} + \frac{1}{27}$ (1); common denominator (1); sum $\frac{7}{108}$ (1).	1+1+1
17	Total $= 7.5 \times 10^3 \times 16$ (1) $= 120 \times 10^3$ (1); in scientific notation 1.2×10^5 (1).	1+1+1
	OR: $5^2 \times 2^3$ (1+1); value $25 \times 8 = 200$ (1).	1+1+1
18	$\frac{1}{100} + \frac{1}{16}$ (1); common denominator (1); sum $\frac{29}{400}$ (1).	1+1+1
	OR: Total $= 8 \times 10^4 \times 125$ (1) $= 1000 \times 10^4$ (1); in scientific notation 1×10^7 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Numerator power of 3: $3^3 \div 3^{-1} = 3^4$ (2); for 4: $4^{-2} \div 4^1 = 4^{-3} = \frac{1}{4^3}$ (1); result $\frac{3^4}{4^3}$ (1); value $\frac{81}{64}$ (1).	2+1+1+1
	OR: Number of cells $= \dfrac{24 \times 10^1}{8 \times 10^{-5}}$ (2); $= \frac{24}{8} \times 10^{1+5}$ (2); 3×10^6 (1).	2+2+1
20	Numerator power of 5: $5^2 \div 5^{-1} = 5^3$ (2); for 3: $3^{-1} \div 3^2 = 3^{-3} = \frac{1}{3^3}$ (1); result $\frac{5^3}{3^3}$ (1); value $\frac{125}{27}$ (1).	2+1+1+1
	OR: Number of cells $= \dfrac{9 \times 10^2}{3 \times 10^{-7}}$ (2); $= \frac{9}{3} \times 10^{2+7}$ (2); 3×10^9 (1).	2+2+1

Mistakes to avoid:

- Writing $2^{-3} = -8$ or $-\frac{1}{8}$; a negative exponent means the **reciprocal**, so $2^{-3} = \frac{1}{8}$.
- Adding the bases: $2^3 \times 2^4 = 4^7$ instead of $2^{3+4} = 2^7$.
- Saying $a^0 = 0$; for $a \neq 0$ it is 1.
- Multiplying exponents when multiplying powers of the same base ($a^m \times a^n = a^{mn}$) or adding them in $(a^m)^n$.
- Leaving the answer as 25×10^4 or 0.25×10^6; in standard form the number part must satisfy $1 \leq m < 10$.
- Getting the sign of the exponent wrong for small numbers: $0.0003 = 3 \times 10^{-4}$, not 3×10^4.

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